

Every Ingredient, Every Step

Twelve T6 Mythic keepers | Raw gathering, processing, vendor supplies and rerolls

Icha has a tremendous plan, uwu. Every ingot, every plank, every rope, every piece of leather - traced back to the resources that make it. This document expands the corrected salvage plan instead of replacing it.

The whole production campaign	Budget
Donor tools / starting tools including 12 keepers	16,151 / 16,163
Finished ingredients used directly in tools	883,400
Additional Rough Planks for bucket production	11,780
Reforging Solvents / filled Water Buckets	3,896 / 117,792
Hex Coins for all listed Rumbagh supplies	7,086,120
Completion lower bound with buffered fish oil	99.024%

The quantities assume existing settlement stations, zero starting stock, ordinary recipes and no bonus output. Whole batches are rounded upward. The fish allowance covers processing the specified species; catching or acquiring those species remains a gathering task.

Public recipe snapshot: September 21, 2026. The cereal/cs branch was checked October 5, 2026 and still points to that snapshot. Pink cuffs, hearts and a very serious shopping campaign.

One Bill, Three Views

Gather the raw list, make the intermediates, then spend the finished ingredients

♥ The columns are different views of one project

+ **Raw shopping lists:** ore chunks, trunk cargo, root cargo, animal carcasses, selected whole fish, catalyst seeds and purchased supplies. These are the acquisition targets.

Production ledgers: how many times each recipe runs, and what those runs produce. These intermediate quantities are already funded by the raw shopping lists. They are not extra purchases.

Finished tool ingredients: four ingots plus two each of rope, planks and leather per tool at each tier. T2 and above also consume the preceding-tier tool, carried forward from this campaign.

♥ The plan Icha is funding

Tier	Tools crafted	Keeper target	Donors scrapped
T1	16,163	Common	-
T2	16,163	Uncommon	148
T3	16,015	Rare	367
T4	15,648	Epic	1,439
T5	14,209	Legendary	4,067
T6	10,142	Mythic	10,130

Donor batches are assigned to their eventual scrap tier before rarity rolls. Every allocated donor is scrapped at that tier, whatever its rarity. Only the twelve keeper tools receive paid rerolls.

Shopping quantities ignore random byproducts and output bonuses. Food, ammunition, equipment replacements, storage and construction are separate operating/setup costs because Icha has not supplied a loadout or a settlement inventory.

Ore for the Entire Campaign

Ordinary route: ore cargo -> ore pieces -> concentrate -> molten metal -> ingots

Tier / metal	Ore chunks	Ore pieces	Concentrate	Ingots
T1 Ferralith	32,326	129,304	64,652	64,652
T2 Pyrelite	32,326	129,304	64,652	64,652
T3 Emarium	64,060	256,240	128,120	64,060
T4 Elenvar	62,592	250,368	125,184	62,592
T5 Luminite	85,254	341,016	170,508	56,836
T6 Rathium	60,852	243,408	121,704	40,568

♥ Why the ore bill rises at higher tiers

Every ore chunk gives **4 ore pieces**. Every **2 pieces** give one concentrate. A molten ingot takes **1 concentrate at T1-T2, 2 at T3-T4, and 3 at T5-T6**. Each molten metal also consumes one matching Wood Log as fuel.

Whole-project mining total	Quantity
Ore chunks across all six metals	337,410
Ore pieces processed	1,349,640
Ore concentrate processed	674,820
Molten metal / finished ingots	353,360

Wood, Bark and Fuel

Trunks are allocated to either logs or bark; the two allocations are added once

Tier	Trunks: logs	Trunks: bark	Total trunks	Logs needed
T1	32,829	2,694	35,523	196,970
T2	26,939	2,694	29,633	161,630
T3	42,707	2,670	45,377	256,240
T4	41,728	2,608	44,336	250,368
T5	52,100	2,369	54,469	312,598
T6	37,188	1,691	38,879	223,124

Three demands share the wood supply

Planks: matching logs are sawn into Stripped Wood, then treated into Planks.

Smelting fuel: one matching log per molten metal.

Tannin: matching bark plus filled water buckets. T1 also supplies all planks used to make buckets for every tier.

Whole-project forestry total	Quantity
Trunks allocated to log production	233,491
Trunks allocated to bark production	14,726
All trunks to acquire	248,217
Matching bark used in tannin	58,896

A trunk produces 6 logs OR 4 bark. This list does not count both outputs from the same trunk. Direct bark drops and existing logs can reduce what Icha needs to acquire.

Fiber and Pelts, Fully Expanded

Roots -> fiber -> thread -> cloth strips -> rope | carcasses -> pelts -> leather

♥ The rope supply

Tier	Root cargo	Fiber needed	Thread spools	Rope
T1	12,931	1,293,040	32,326	32,326
T2	12,931	1,293,040	32,326	32,326
T3	25,624	2,562,400	64,060	32,030
T4	25,037	2,503,680	62,592	31,296
T5	34,102	3,410,160	85,254	28,418
T6	24,341	2,434,080	60,852	20,284

♥ The leather supply

Tier	Carcasses	Raw pelts	Tanned pelts	Leather
T1	10,776	32,326	32,326	32,326
T2	10,776	32,326	32,326	32,326
T3	21,354	64,060	32,030	32,030
T4	20,864	62,592	31,296	31,296
T5	28,418	85,254	28,418	28,418
T6	20,284	60,852	20,284	20,284

Roots give 100 matching Plant Fiber each; 40 fiber makes one thread spool. The chosen ordinary hunting recipes give 3 matching Raw Pelts per carcass. Species and cargo names appear on the tier shopping pages.

Vendor Supplies and Hex Coins

All four purchased ingredients | Prices verified against traveler trade orders

Purchase	Quantity	Coins each	Coin subtotal
Metalworking Flux	424,620	10	4,246,200
Woodworking Sandpaper	212,310	2	424,620
Hideworking Salt	212,310	10	2,123,100
Reforging Stabilizer	3,896	75	292,200
TOTAL			7,086,120

The coin budget by production tier

Tier	Flux	Sandpaper	Salt	Total coins
T1	0	0	0	0
T2	0	0	0	4,950
T3	64,060	32,030	32,030	1,036,360
T4	125,184	62,592	62,592	2,027,094
T5	113,672	56,836	56,836	1,868,552
T6	121,704	60,852	60,852	2,149,164

Tier coin totals include that tier's Reforging Stabilizers. The earlier 292,200-coin total covered stabilizers alone; it did not cover flux, salt or sandpaper. This page funds all four.

Water Is a Production Chain

Centralized T1 bucket production; already included in the T1 raw-material list

Operation	Runs	Guaranteed result
Saw 3 Rough Logs into 1 Rough Stripped Wood	11,780	11,780 Stripped Wood
Craft 1 Rough Stripped Wood into 1 Rough Plank	11,780	11,780 Planks
Assemble 1 Rough Plank into 10 Empty Buckets	11,780	117,800 Empty Buckets
Fill 1 Empty Bucket at a Well	117,792	117,792 Water Buckets

Where the water goes

Tier	Tannin batches	Tannin produced	Water Buckets
T1	10,776	32,328	21,552
T2	10,776	32,328	21,552
T3	10,677	32,031	21,354
T4	10,432	31,296	20,864
T5	9,473	28,419	18,946
T6	6,762	20,286	13,524

One batch uses 1 matching Tree Bark + 2 Water Buckets and makes 3 Tannin. Icha makes 11,780 bucket batches once for the whole plan; 8 empty buckets remain. The selected recipes consume filled buckets and list no empty-bucket return.

The Metal and Plank Chains

All quantities are for one guaranteed finished unit unless a batch output is shown

♥ Metal, from cargo to ingot

Step	Recipe
Crush ore	1 Ore Chunk -> 4 matching Ore Pieces
Make concentrate	2 Ore Pieces -> 1 matching Ore Concentrate
Smelt	1 / 1 / 2 / 2 / 3 / 3 Concentrate + 1 matching Wood Log -> 1 Molten Metal, T1 through T6
Forge	1 Molten Metal + 0 / 0 / 1 / 2 / 2 / 3 Metalworking Flux -> 1 Ingot

♥ Wood, from trunk to finished plank

Step	Recipe
Split logs	1 matching Trunk -> 6 Wood Logs
Saw stripped wood	3 matching Wood Logs -> 1 Stripped Wood
Treat planks	1 / 1 / 2 / 2 / 3 / 3 Stripped Wood + 0 / 0 / 1 / 2 / 2 / 3 Sandpaper -> 1 Plank, T1 through T6
Split bark instead	1 matching Trunk -> 4 Tree Bark; separate from the log recipe

This plan uses the ordinary one-plank recipe. Hexite Wood Fragment recipes can double plank output, but those are an optional substitution described on page 24.

The Rope and Leather Chains

No unexplained finished-material shortcuts: each intermediate is shown

♥ Rope, from root cargo

Step	Recipe
Extract fiber	1 matching Plant Roots cargo -> 100 Plant Fiber
Spin thread	40 matching Plant Fiber -> 1 Spool Of Thread
Weave cloth strip	1 / 1 / 2 / 2 / 3 / 3 matching thread spools -> 1 Cloth Strip, T1 through T6
Braid rope	1 matching Cloth Strip -> 1 Rope

♥ Leather, from a hunted carcass

Step	Recipe
Harvest carcass	1 selected animal carcass -> 3 matching Raw Pelts, plus meat
Clean	1 Raw Pelt -> 1 matching Cleaned Pelt
Make tannin	1 matching Tree Bark + 2 Water Buckets -> 3 matching Tannin
Tan	1 / 1 / 2 / 2 / 3 / 3 Cleaned Pelts + 1 Tannin -> 1 Tanned Pelt, T1 through T6
Stitch leather	1 matching Tanned Pelt + 0 / 0 / 1 / 2 / 2 / 3 Hideworking Salt -> 1 Leather

T3 uses Sturdy wood/pelts/fiber but Infused tannin, oil, seeds, catalyst and solvent. These are the actual item grades in the recipes.

T1 Raw Shopping List

Rough / Ferralith | 16,163 tools at this tier | Twelve keepers included

Tool ingredients: 64,652 Ingots; 32,326 Rope; 32,326 Planks; 32,326 Leather. Plus 11,780 Rough Planks for the shared bucket factory.

Done	Gather or acquire	Quantity
☐	Ferralith Ore Chunk	32,326
☐	Rough Wood Trunk	35,523
☐	Rough Plant Roots	12,931
☐	Sagi Bird	10,776

♥ Buy from Rumbagh

No Rumbagh consumables are required for T1 in this route.

Tier coin budget: 0 Hex Coins.

Trunk allocation: 32,829 to logs + 2,694 to bark. The raw list already adds these together.

Roots can be replaced by 1,293,040 already gathered Rough Plant Fiber. Carcasses can be replaced by 32,326 Rough Raw Pelts. These are substitutions, not additions.

All tiers' bucket-making wood is charged here, so later tiers do not purchase bucket wood again.

T1 Every Crafting Step

Runs are recipe batches; outputs are intermediates already funded by the raw checklist

Production step	Runs	Output	Recipe ID
Crush ore	32,326	129,304 Ore Pieces	103005
Make concentrate	64,652	64,652 Concentrate	103006
Smelt with log fuel	64,652	64,652 Molten Ferralith	105000
Forge ingots	64,652	64,652 Ingots	105009
Split trunks to logs	32,829	196,974 Wood Logs	101003
Saw stripped wood	44,106	44,106 Stripped Wood	102000
Treat planks	44,106	44,106 Planks	102009
Split trunks to bark	2,694	10,776 Tree Bark	101004
Extract root fiber	12,931	1,293,100 Plant Fiber	1517386067
Spin thread	32,326	32,326 Thread Spools	109007
Weave cloth strips	32,326	32,326 Cloth Strips	109000
Braid rope	32,326	32,326 Rope	109005
Harvest carcasses	10,776	32,328 Raw Pelts	106000
Clean pelts	32,326	32,326 Cleaned Pelts	107004
Mix tannin	10,776	32,328 Tannin	107010
Tan pelts	32,326	32,326 Tanned Pelts	107011
Stitch leather	32,326	32,326 Leather	667502937

The T1 plank total includes 11,780 bucket planks. Run bucket assembly recipe 1000 that many times, then fill 117,792 buckets using recipe 1230133585. See page 7.

Recipe IDs refer to the pinned crafting_recipe_desc.json source linked on page 26. Integer batch surpluses are deliberately left unused.

T2 Raw Shopping List

Simple / Pyrelite | 16,163 tools at this tier | Twelve keepers included

Tool ingredients: 64,652 Ingots; 32,326 Rope; 32,326 Planks; 32,326 Leather.

Done	Gather or acquire	Quantity
☐	Pyrelite Ore Chunk	32,326
☐	Simple Wood Trunk	29,633
☐	Simple Plant Roots	12,931
☐	Nubi Goat	10,776
☐	Muddy Oncor	44
☐	Simple Wispweave Seeds	660

♥ Buy from Rumbagh

Done	Purchased ingredient	Quantity
☐	Reforging Stabilizer	66

Tier coin budget: 4,950 Hex Coins.

Trunk allocation: 26,939 to logs + 2,694 to bark. The raw list already adds these together.

Roots can be replaced by 1,293,040 already gathered Simple Plant Fiber. Carcasses can be replaced by 32,326 Simple Raw Pelts. These are substitutions, not additions.

The fish quantity is a buffered processing budget for the named species. Stop once 132 Simple Fish Oil is secured. Catalyst seeds may be replaced by the same quantity of matching Embergrain or Starbulb seeds.

T2 Every Crafting Step

Runs are recipe batches; outputs are intermediates already funded by the raw checklist

Production step	Runs	Output	Recipe ID
Crush ore	32,326	129,304 Ore Pieces	203005
Make concentrate	64,652	64,652 Concentrate	203006
Smelt with log fuel	64,652	64,652 Molten Pyrelite	205000
Forge ingots	64,652	64,652 Ingots	205009
Split trunks to logs	26,939	161,634 Wood Logs	201003
Saw stripped wood	32,326	32,326 Stripped Wood	1624893582
Treat planks	32,326	32,326 Planks	202009
Split trunks to bark	2,694	10,776 Tree Bark	201004
Extract root fiber	12,931	1,293,100 Plant Fiber	56062210
Spin thread	32,326	32,326 Thread Spools	209007
Weave cloth strips	32,326	32,326 Cloth Strips	1376564210
Braid rope	32,326	32,326 Rope	209005
Harvest carcasses	10,776	32,328 Raw Pelts	206000
Clean pelts	32,326	32,326 Cleaned Pelts	207004
Mix tannin	10,776	32,328 Tannin	207010
Tan pelts	32,326	32,326 Tanned Pelts	1068965515
Stitch leather	32,326	32,326 Leather	207011
Process selected fish	44	132 Oil target *	211011
Craft catalysts	66	66 Catalysts	1467169816
Craft solvents	66	66 Solvents	1260268122

Each catalyst uses 2 matching Fish Oil + 10 matching Wispweave Seeds. Each solvent uses 1 matching Catalyst + 1 Reforging Stabilizer. * Oil output is probabilistic; the shopping count includes a separate tail-risk allowance.

Keeper reroll cap: 66 paid rerolls; 198 matching scrap; 66 solvents. Scrap 148 allocated donors. The initial free keeper upgrade roll is included in the probability model.

T3 Raw Shopping List

Sturdy / Emarium | 16,015 tools at this tier | Twelve keepers included

Tool ingredients: 64,060 Ingots; 32,030 Rope; 32,030 Planks; 32,030 Leather.

Done	Gather or acquire	Quantity
☐	Emarium Ore Chunk	64,060
☐	Sturdy Wood Trunk	45,377
☐	Sturdy Plant Roots	25,624
☐	Female Cervus	21,354
☐	Greenhorn Gorbu	266
☐	Infused Wispweave Seeds	1,520

♥ Buy from Rumbagh

Done	Purchased ingredient	Quantity
☐	Metalworking Flux	64,060
☐	Woodworking Sandpaper	32,030
☐	Hideworking Salt	32,030
☐	Reforging Stabilizer	152

Tier coin budget: 1,036,360 Hex Coins.

Trunk allocation: 42,707 to logs + 2,670 to bark. The raw list already adds these together.

Roots can be replaced by 2,562,400 already gathered Sturdy Plant Fiber. Carcasses can be replaced by 64,060 Sturdy Raw Pelts. These are substitutions, not additions.

The fish quantity is a buffered processing budget for the named species. Stop once 304 Infused Fish Oil is secured. Catalyst seeds may be replaced by the same quantity of matching Embergrain or Starbulb seeds.

T3 Every Crafting Step

Runs are recipe batches; outputs are intermediates already funded by the raw checklist

Production step	Runs	Output	Recipe ID
Crush ore	64,060	256,240 Ore Pieces	303005
Make concentrate	128,120	128,120 Concentrate	303006
Smelt with log fuel	64,060	64,060 Molten Emarium	305000
Forge ingots	64,060	64,060 Ingots	305009
Split trunks to logs	42,707	256,242 Wood Logs	301003
Saw stripped wood	64,060	64,060 Stripped Wood	1859292277
Treat planks	32,030	32,030 Planks	302009
Split trunks to bark	2,670	10,680 Tree Bark	301004
Extract root fiber	25,624	2,562,400 Plant Fiber	1926980078
Spin thread	64,060	64,060 Thread Spools	309007
Weave cloth strips	32,030	32,030 Cloth Strips	1801327855
Braid rope	32,030	32,030 Rope	309005
Harvest carcasses	21,354	64,062 Raw Pelts	306000
Clean pelts	64,060	64,060 Cleaned Pelts	307004
Mix tannin	10,677	32,031 Tannin	307010
Tan pelts	32,030	32,030 Tanned Pelts	982803753
Stitch leather	32,030	32,030 Leather	307011
Process selected fish	266	304 Oil target *	311011
Craft catalysts	152	152 Catalysts	1133765659
Craft solvents	152	152 Solvents	1321911790

Each catalyst uses 2 matching Fish Oil + 10 matching Wispweave Seeds. Each solvent uses 1 matching Catalyst + 1 Reforging Stabilizer. * Oil output is probabilistic; the shopping count includes a separate tail-risk allowance.

Keeper reroll cap: 152 paid rerolls; 760 matching scrap; 152 solvents. Scrap 367 allocated donors. The initial free keeper upgrade roll is included in the probability model.

T4 Raw Shopping List

Fine / Elenvar | 15,648 tools at this tier | Twelve keepers included

Tool ingredients: 62,592 Ingots; 31,296 Rope; 31,296 Planks; 31,296 Leather.

Done	Gather or acquire	Quantity
☐	Elenvar Ore Chunk	62,592
☐	Fine Trunk	44,336
☐	Fine Plant Roots	25,037
☐	Rangifer	20,864
☐	Azure Centro	720
☐	Fine Wispweave Seeds	3,220

♥ Buy from Rumbagh

Done	Purchased ingredient	Quantity
☐	Metalworking Flux	125,184
☐	Woodworking Sandpaper	62,592
☐	Hideworking Salt	62,592
☐	Reforging Stabilizer	322

Tier coin budget: 2,027,094 Hex Coins.

Trunk allocation: 41,728 to logs + 2,608 to bark. The raw list already adds these together.

Roots can be replaced by 2,503,680 already gathered Fine Plant Fiber. Carcasses can be replaced by 62,592 Fine Raw Pelts. These are substitutions, not additions.

The fish quantity is a buffered processing budget for the named species. Stop once 644 Fine Fish Oil is secured. Catalyst seeds may be replaced by the same quantity of matching Embergrain or Starbulb seeds.

T4 Every Crafting Step

Runs are recipe batches; outputs are intermediates already funded by the raw checklist

Production step	Runs	Output	Recipe ID
Crush ore	62,592	250,368 Ore Pieces	403005
Make concentrate	125,184	125,184 Concentrate	403006
Smelt with log fuel	62,592	62,592 Molten Elenvar	405000
Forge ingots	62,592	62,592 Ingots	405009
Split trunks to logs	41,728	250,368 Wood Logs	401003
Saw stripped wood	62,592	62,592 Stripped Wood	1529267053
Treat planks	31,296	31,296 Planks	402009
Split trunks to bark	2,608	10,432 Tree Bark	401004
Extract root fiber	25,037	2,503,700 Plant Fiber	807480458
Spin thread	62,592	62,592 Thread Spools	409007
Weave cloth strips	31,296	31,296 Cloth Strips	1185486224
Braid rope	31,296	31,296 Rope	409005
Harvest carcasses	20,864	62,592 Raw Pelts	406000
Clean pelts	62,592	62,592 Cleaned Pelts	407004
Mix tannin	10,432	31,296 Tannin	407010
Tan pelts	31,296	31,296 Tanned Pelts	1725687319
Stitch leather	31,296	31,296 Leather	407011
Process selected fish	720	644 Oil target *	411011
Craft catalysts	322	322 Catalysts	2130933601
Craft solvents	322	322 Solvents	1864168115

Each catalyst uses 2 matching Fish Oil + 10 matching Wispweave Seeds. Each solvent uses 1 matching Catalyst + 1 Reforging Stabilizer. * Oil output is probabilistic; the shopping count includes a separate tail-risk allowance.

Keeper reroll cap: 322 paid rerolls; 4,830 matching scrap; 322 solvents. Scrap 1,439 allocated donors. The initial free keeper upgrade roll is included in the probability model.

T5 Raw Shopping List

Exquisite / Luminite | 14,209 tools at this tier | Twelve keepers included

Tool ingredients: 56,836 Ingots; 28,418 Rope; 28,418 Planks; 28,418 Leather.

Done	Gather or acquire	Quantity
☐	Luminite Ore Chunk	85,254
☐	Exquisite Trunk	54,469
☐	Exquisite Plant Roots	34,102
☐	Plains Ox	28,418
☐	Rocky Aulono	2,612
☐	Exquisite Wispweave Seeds	6,640

♥ Buy from Rumbagh

Done	Purchased ingredient	Quantity
☐	Metalworking Flux	113,672
☐	Woodworking Sandpaper	56,836
☐	Hideworking Salt	56,836
☐	Reforging Stabilizer	664

Tier coin budget: 1,868,552 Hex Coins.

Trunk allocation: 52,100 to logs + 2,369 to bark. The raw list already adds these together.

Roots can be replaced by 3,410,160 already gathered Exquisite Plant Fiber. Carcasses can be replaced by 85,254 Exquisite Raw Pelts. These are substitutions, not additions.

The fish quantity is a buffered processing budget for the named species. Stop once 1,328 Exquisite Fish Oil is secured. Catalyst seeds may be replaced by the same quantity of matching Embergrain or Starbulb seeds.

T5 Every Crafting Step

Runs are recipe batches; outputs are intermediates already funded by the raw checklist

Production step	Runs	Output	Recipe ID
Crush ore	85,254	341,016 Ore Pieces	503005
Make concentrate	170,508	170,508 Concentrate	503006
Smelt with log fuel	56,836	56,836 Molten Luminite	505000
Forge ingots	56,836	56,836 Ingots	505009
Split trunks to logs	52,100	312,600 Wood Logs	501003
Saw stripped wood	85,254	85,254 Stripped Wood	2079251095
Treat planks	28,418	28,418 Planks	502009
Split trunks to bark	2,369	9,476 Tree Bark	501004
Extract root fiber	34,102	3,410,200 Plant Fiber	1640720421
Spin thread	85,254	85,254 Thread Spools	509007
Weave cloth strips	28,418	28,418 Cloth Strips	1285503944
Braid rope	28,418	28,418 Rope	509005
Harvest carcasses	28,418	85,254 Raw Pelts	506000
Clean pelts	85,254	85,254 Cleaned Pelts	507004
Mix tannin	9,473	28,419 Tannin	507010
Tan pelts	28,418	28,418 Tanned Pelts	1282160349
Stitch leather	28,418	28,418 Leather	507011
Process selected fish	2,612	1,328 Oil target *	511011
Craft catalysts	664	664 Catalysts	1472654957
Craft solvents	664	664 Solvents	810317042

Each catalyst uses 2 matching Fish Oil + 10 matching Wispweave Seeds. Each solvent uses 1 matching Catalyst + 1 Reforging Stabilizer. * Oil output is probabilistic; the shopping count includes a separate tail-risk allowance.

Keeper reroll cap: 664 paid rerolls; 19,920 matching scrap; 664 solvents. Scrap 4,067 allocated donors. The initial free keeper upgrade roll is included in the probability model.

T6 Raw Shopping List

Peerless / Rathium | 10,142 tools at this tier | Twelve keepers included

Tool ingredients: 40,568 Ingots; 20,284 Rope; 20,284 Planks; 20,284 Leather.

Done	Gather or acquire	Quantity
☐	Rathium Ore Chunk	60,852
☐	Peerless Trunk	38,879
☐	Peerless Plant Roots	24,341
☐	Yagi	20,284
☐	Vibrant Janus	10,180
☐	Peerless Wispweave Seeds	26,920

♥ Buy from Rumbagh

Done	Purchased ingredient	Quantity
☐	Metalworking Flux	121,704
☐	Woodworking Sandpaper	60,852
☐	Hideworking Salt	60,852
☐	Reforging Stabilizer	2,692

Tier coin budget: 2,149,164 Hex Coins.

Trunk allocation: 37,188 to logs + 1,691 to bark. The raw list already adds these together.

Roots can be replaced by 2,434,080 already gathered Peerless Plant Fiber. Carcasses can be replaced by 60,852 Peerless Raw Pelts. These are substitutions, not additions.

The fish quantity is a buffered processing budget for the named species. Stop once 5,384 Peerless Fish Oil is secured. Catalyst seeds may be replaced by the same quantity of matching Embergrain or Starbulb seeds.

T6 Every Crafting Step

Runs are recipe batches; outputs are intermediates already funded by the raw checklist

Production step	Runs	Output	Recipe ID
Crush ore	60,852	243,408 Ore Pieces	603005
Make concentrate	121,704	121,704 Concentrate	603006
Smelt with log fuel	40,568	40,568 Molten Rathium	605000
Forge ingots	40,568	40,568 Ingots	605009
Split trunks to logs	37,188	223,128 Wood Logs	601003
Saw stripped wood	60,852	60,852 Stripped Wood	1423759589
Treat planks	20,284	20,284 Planks	602009
Split trunks to bark	1,691	6,764 Tree Bark	601004
Extract root fiber	24,341	2,434,100 Plant Fiber	1504494184
Spin thread	60,852	60,852 Thread Spools	609007
Weave cloth strips	20,284	20,284 Cloth Strips	2075880051
Braid rope	20,284	20,284 Rope	609005
Harvest carcasses	20,284	60,852 Raw Pelts	606000
Clean pelts	60,852	60,852 Cleaned Pelts	607004
Mix tannin	6,762	20,286 Tannin	607010
Tan pelts	20,284	20,284 Tanned Pelts	1260921480
Stitch leather	20,284	20,284 Leather	607011
Process selected fish	10,180	5,384 Oil target *	611011
Craft catalysts	2,692	2,692 Catalysts	800950956
Craft solvents	2,692	2,692 Solvents	1887033805

Each catalyst uses 2 matching Fish Oil + 10 matching Wispweave Seeds. Each solvent uses 1 matching Catalyst + 1 Reforging Stabilizer. * Oil output is probabilistic; the shopping count includes a separate tail-risk allowance.

Keeper reroll cap: 1,346 paid rerolls; 67,300 matching scrap; 2,692 solvents. Scrap 10,130 allocated donors. The initial free keeper upgrade roll is included in the probability model.

The Fish Are Not All Equal

The plan selects ordinary oceanfish with better oil yields than the paired inland fish

Tier / selected fish	Oil needed	Mean oil / fish	Fish budget
T2 Muddy Oncor	132	3.05	44
T3 Greenhorn Gorbu	304	1.55	266
T4 Azure Centro	644	1.05	720
T5 Rocky Aulono	1,328	0.55	2,612
T6 Vibrant Janus	5,384	0.55	10,180

♥ What the selected processing recipes roll

Fish	Oil outputs and probabilities
Muddy Oncor	3 oil: 95%; 4 oil: 5%
Greenhorn Gorbu	0 oil: 50%; 3 oil: 45%; 4 oil: 5%
Azure Centro	0 oil: 45%; 1 oil: 5%; 2 oil: 50%
Rocky Aulono	0 oil: 45%; 1 oil: 55%
Vibrant Janus	0 oil: 45%; 1 oil: 55%

T2 needs only 44 Muddy Oncor because each gives at least 3 Simple Fish Oil. For T3-T6, each fixed fish budget has a proven oil-shortfall bound below 0.01%. These are counts of the named whole fish ready for processing, not total fishing attempts or all mixed catches.

Icha may buy matching oil directly instead: 132 Simple, 304 Infused, 644 Fine, 1,328 Exquisite and 5,384 Peerless. Then remove the corresponding whole-fish budget rather than adding oil on top.

Stations Icha Needs to Access

Existing settlement stations assumed; construction is not charged to these production totals

Station	Access tier	Used for
Mining Station	T6	Ore cargo -> ore pieces
Smithing Station	T6	Concentrate, ingots, tools and upgrades
Smelter	T6	Molten metal; fuel logs included
Forestry Station	T6	Trunks -> logs or bark
Carpentry Station	T6	Stripped wood, planks; T1 buckets
Foraging Station	T6	Roots -> fiber
Tailoring Station	T6	Thread and rope
Loom	T6	Thread -> cloth strips
Hunting Station	T6	Carcasses -> raw pelts
Leatherworking Station	T6	Cleaning, tannin and leather
Tanning Tub	T6	Cleaned pelts -> tanned pelts
Fishing Station	T6	Whole fish -> oil
Scholar Station	T6	Oil + seeds -> chemical catalysts
Scrap Bench	T1	Reforging Solvents; scrap and rerolls
Well	T1	Empty Bucket -> Water Bucket

The tier-specific recipes require the matching station tier; this list shows the highest needed access. A settlement must also meet the recipe's skill and knowledge requirements. Main production professions reach level 60 for T6 recipes; passive recipes may have lower personal-level requirements.

Food, arrows/ammunition, farming inputs if Icha chooses cultivation, gear wear, repairs, transport and building construction depend on the chosen gathering method and existing setup. They are not fixed inputs to the selected production recipes, so no invented quantities are assigned.

Ways to Reduce the Shopping Bill

These routes replace part of the main plan; they are not included in its totals

♥ Hexite planks: two planks instead of one

Tier	Extra fragments / batch	Stripped wood	Sandpaper
T1	2	1	0
T2	3	1	0
T3	4	2	1
T4	5	2	2
T5	6	3	2
T6	7	3	3

Each Hexite recipe makes 2 matching Planks from the same ordinary stripped-wood and sandpaper inputs, plus the fragments shown. Run half as many plank batches for even plank targets. Recalculate upstream logs and vendor sandpaper before substituting; do not keep both complete lists.

♥ Use what Icha already has

Existing fiber: replace root cargo with the listed matching fiber requirement.

Existing bark or logs: deduct them before rounding the corresponding trunk allocation.

Existing pelts: replace carcasses with matching raw pelts.

Existing water buckets or oil: reduce the bucket factory or whole-fish budget.

Existing finished materials: remove that finished quantity and its upstream inputs.

This baseline takes the ordinary pelt-to-leather route, including T1. Refined Rough Leather splitting, event salvage, byproducts and market trades may be useful alternatives, but are not silently mixed into the baseline.

How the Campaign Completes

A source-backed production bill plus the corrected keeper and donor RNG model

♥ The order of work

- + 1. Icha acquires the raw cargo, seeds and named fish and buys the vendor supplies.
2. Icha makes all shared buckets at T1 and fills them at a Well.
3. Icha processes each tier's mining, forestry, textile and hide chains.
4. Icha secures the fish-oil targets, then crafts catalysts and solvents.
5. Icha crafts the T1 tools, carries the assigned batches forward, scraps donors at their assigned tier, and rerolls only keepers.
6. Icha stops each keeper at the highest available rarity before advancing it.

♥ What the 99% statement covers

The inherited keeper/donor model has a completion lower bound of **99.061%**. The four probabilistic fish-oil batches add a combined shortfall bound of **0.0369%**. Subtracting those risks gives a conservative combined lower bound of **99.024%**, conditional on acquiring the listed raw inputs and executing the recipes.



Keeper caps use negative-binomial probabilities. Donor salvage and fish-oil caps use lower-tail Chernoff bounds from their actual output distributions. The union bound does not require those failure events to be independent. Recipe rolls within each batch follow the published model.

♥ What was checked

Every selected recipe's inputs and guaranteed outputs were asserted against the pinned JSON. An independent aggregate conservation check executed the recipe batches, deducted all finished tool and solvent requirements, and found no material deficits. Whole-batch surpluses remain unused. No extra credit was assumed for crits, bonus output or random byproducts.

The Evidence Behind the Bill

Pinned public data | Source IDs printed in every tier production ledger

♥ Published tables used

+ **crafting_recipe_desc.json**

Craft input quantities, output batches, stations and recipe IDs

item_list_desc.json

Guaranteed log/bark/pelt outputs, bucket batches and fish-oil probabilities

♥ **traveler_trade_order_desc.json**

Rumbagh prices: flux 10, sandpaper 2, salt 10 and stabilizers 75 Hex Coins

item_desc.json

Actual item names, tiers, chemical grades and output-list references

cargo_desc.json

Ore chunks, trunk cargo, roots and animal carcass names

building_type_desc.json

Names of required crafting stations

♥ Snapshot and model

Public cereal/cs commit:

5b25073bad925972b0e7543172f3c2d445f58f5a

Snapshot dated September 21, 2026; branch revision verified October 5, 2026.

The twelve-tool donor/roll quantities come from the corrected actual-salvage plan. This document adds the entire selected production chain, shared bucket demand, raw hunting and foraging routes, vendor supplies and fish-oil allowances. Original PDFs remain separate.

Coverage is the complete consumable production bill for the specified recipe route using existing stations. Gathering attempts, time, food, combat ammunition and settlement construction require player-specific assumptions and are not represented as fixed recipe inputs.